

Conor Knowles

Software Development Engineer

📍 Dublin 🌐 @conorhk 🌐 @conorhknowles ✉ censored 📞 censored

EXPERIENCE

AMAZON WEB SERVICES | SOFTWARE DEVELOPMENT ENGINEER II

Sept 2022 – Current | Dublin, Ireland

- **Led a team of 6 backend engineers** to migrate legacy data transfer objects to a modern framework to mitigate future failure points. The new framework was implemented across distributed services with zero operational impact.
- **Developed advanced patch management workflow for internal customer-carrying hosts** with dynamic load balancing, automated rollback capabilities, and zero-to-hero host provisioning. The workflow utilized CDK-based CI/CD pipelines to deploy across multiple AWS regions. The solution automates what was previously 6 weeks of effort.
- Implemented a unified device operation endpoint using **Python**, consolidating access to 188 existing APIs across multiple vendor ecosystems. **Evaluated and designed API interfaces to provide vendor-agnostic device access** while maintaining strict security controls and operational safety. The solution standardized device operations and reduced integration complexity for network automation workflows and enabled critical network scaling efforts.
- Designed a network topology validation system to mitigate availability drops and resource load for a **tier-1 ECS service**, reducing the maintenance effort for the service by 80%.

AMAZON WEB SERVICES | SYSTEMS DEVELOPMENT ENGINEER INTERN

May 2020 – Aug 2021 | Dublin, Ireland

- **Developed a service integration to generate software configurations for commodity hardware in a zero-touch-provisioning ecosystem.** The solution is used to automatically deploy 40,000 commodity devices a year.

ACADEMY OF CODE | PROGRAMMING TEACHER

Sept 2020 – March 2021 | Dublin, Ireland

- **Taught Java programming to 50+ students aged 6-17**, developing age-appropriate curriculum and hands-on learning experiences that improved student engagement and programming fundamentals comprehension.

PROJECTS

HOMELAB | NIX, DOCKER, KUBERNETES

2022 - Current

- **Architected and deployed a homelab cluster using NixOS, Docker, and Kubernetes** using leftover hardware, hosting 20+ services including reverse proxy, DNS server, AI model, and family productivity applications. Implemented secure remote access via a mesh VPN network using WireGuard, providing frictionless security and reliability. Serves 12 users across 3 countries.

CUSTOM-BUILT DEVELOPMENT ENVIRONMENT | LUA, VIMSCRIPT, LINUX, BASH

2019 - Current

- Developed advanced NeoVim distribution using Lua and Vimscript integrated with Nix home-manager for reproducible development environment deployment. The distribution provides a **consistent reproducible development environment for 6+ engineers**.

SKILLS

PROGRAMMING

Proficient:

Python • Bash • Java

Experienced:

Lua • JavaScript • C

ENGINEERING

- Software design
- Infrastructure design
- Networking at scale
- DevOps

TOOLS

- Infrastructure as Code
- Linux
- Docker
- AWS
- Nix

EDUCATION

UNIVERSITY COLLEGE DUBLIN

BACHELOR'S IN COMPUTER SCIENCE

Sept 2018 - May 2022 | UCD, Ireland

School of Computing

First Class Honours

CENSORED

LEAVING CERTIFICATE

Sept 2012 - June 2018 | Clare, Ireland

544 points

EXTRA

KAYAKING

Level 3 Whitewater Kayaker

Level 2 Kayak Instructor

KARATE

Black belt

2016 - World Shotokan Karate Federation

Represented Ireland in British

Open

2015 - Glasgow

FIRST AID

REC 3 Rescue First Aider

AWS Office First Aider